Sample Paper 1 — Answer Key

Total: 40 marks (30 × 1 mark + 5 × 2 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Sphere** — the other three are flat (2-D) shapes; a sphere is 3-D.
2. **B) 5** — a pentagon has 5 sides.
3. **A) EPH** — the rule is "+1 letter": D→E, O→P, G→H.
4. **C) Tanvi** — order is Tanvi → Rohan → Simmi, so Tanvi is first.
5. **C) East** — the 5 m North and 5 m South cancel out; only 3 m East remains.
6. **C) 162** — each term is ×3: 2, 6, 18, 54, 162.
7. **B) Saturday** — 10 days = 7 + 3; after a full week it's Wednesday again, then +3 more days → Saturday.
8. **B) G** — A(1) B(2) C(3) D(4) E(5) F(6) G(7).
9. **D) 42** — 16, 25, 36 are perfect squares (4^2, 5^2, 6^2); 42 is not.
10. **B) 12** — each row's two addends rise by 1 and the sum rises by 2: 8, 10, 12.

Section 2 — Mathematical Reasoning

11. **A) 6245** — 3456 + 2789 = 6245.
12. **A) 4546** — 8002 − 3456 = 4546.
13. **A) 1404** — 234 × 6 = 1404.
14. **B) 117** — 936 ÷ 8 = 117.
15. **A) 4/6** — 4/6 simplifies to 2/3.
16. **A) 3/4** — 3/4 = 0.75, 2/3 ≈ 0.667, so 3/4 is greater.
17. **B) 3250 m** — 3 km = 3000 m; 3000 + 250 = 3250 m.
18. **B) 7 kg 250 g** — 500 g + 750 g = 1250 g = 1 kg 250 g; 4 kg + 2 kg + 1 kg 250 g = 7 kg 250 g.
19. **C) 4** — every corner of a square is a right angle.
20. **B) 38 cm** — Perimeter = 2 × (12 + 7) = 2 × 19 = 38 cm.

Section 3 — Everyday Mathematics

21. **A) ₹340** — 500 − 125 − 35 = 340.
22. **B) 270** — 45 × 6 = 270.

23. **B) 57 boxes, 0 left** — 456 ÷ 8 = 57 exactly.
24. **B) 6 h 30 min** — 8:15 a.m. to 2:15 p.m. is 6 hours; plus 30 more minutes to 2:45 p.m.
25. **B) 1 L 250 mL** — 250 mL × 5 = 1250 mL = 1 L 250 mL.
26. **B) ₹151** — (3 × 45) + (2 × 8) = 135 + 16 = 151.
27. **C) 90 L** — 1/4 of 120 L = 30 L used; 120 − 30 = 90 L left.
28. **A) 59 kg** — 32 + 27 = 59.
29. **B) 180** — 15 × 12 = 180.
30. **B) 27°C** — 18 + 9 = 27.

Section 4 — Achievers Section (2 marks each)

31. **B) 165** — Number = divisor × quotient + remainder = (7 × 23) + 4 = 161 + 4 = 165.
32. **A) 300** — Other number = 1250 − 475 = 775; difference = 775 − 475 = 300.
33. **C) ₹375** — Half of 1000 = 500 spent, 500 left; 1/4 of 500 = 125 spent on books; 500 − 125 = 375 left.
34. **C) 65** — Bhanu = 12, Aditi = 2 × 12 = 24, Chirag = 24 + 5 = 29; total = 12 + 24 + 29 = 65.
35. **B) 19** — Each new joined square adds 3 matchsticks to the previous total: for n squares, matchsticks = 3n + 1. For n = 6: (3 × 6) + 1 = 19.

Scoring guide (out of 40): 32+ is Level-2 cutoff range — strong. 24–31 is solid Level-1 performance — review the missed sections. Below 24 — go back to the relevant week(s) in `docs/03-practice-plan.md` before attempting Sample Paper 2.
